Assignment 3: Air Quality Risk Assessment

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Overview

This assignment simulates one year of PM2.5 air quality monitoring data for an urban monitoring station. The analysis classifies daily pollution risk levels using EPA-inspired thresholds, tracks unhealthy pollution streaks, summarizes seasonal patterns using for loops, and visualizes air quality trends across the year.

Load Packages

```r
library(tidyverse)
library(ggplot2)
```

Part 1: Build the Dataset

Why We Chose This Distribution

We used sample() to randomly select 365 PM2.5 values from a range of 2 to 50 µg/m³. We chose this range because typical urban PM2.5 levels often fall between about 5 and 25 µg/m³ according to EPA air quality guidance, but concentrations can become much higher during events such as wildfires, traffic congestion, or atmospheric inversions. Using values between 2 and 50 µg/m³ allowed us to generate realistic variation while still producing enough unhealthy days to observe meaningful patterns in the analysis.

Create the Dataset

```r
set.seed(262)

# create a sequence of PM2.5 values to sample from
possible_pm25 <- seq(from = 2, to = 50, by = 0.1)

# sample 365 days worth of readings
pm25_readings <- sample(possible_pm25, size = 365, replace = TRUE)

# create data frame
air_quality <- data.frame(
  day = 1:365,
  pm25 = pm25_readings
)

# summary statistics
summary(air_quality$pm25)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
   2.20   15.80   27.20   26.96   39.10   50.00 
```

Assign Seasons

```r
# assign season based on day number
# days 1-90 = winter
# days 91-180 = spring
# days 181-270 = summer
# days 271-365 = fall

air_quality$season <- ifelse(
  air_quality$day <= 90, "winter",
  ifelse(
    air_quality$day <= 180, "spring",
    ifelse(air_quality$day <= 270, "summer", "fall")
  )
)

# convert to factor
air_quality$season <- factor(
  air_quality$season,
```

```r
  levels = c("winter", "spring", "summer", "fall")
)

# check results
summary(air_quality$season)
```

```
winter spring summer   fall 
    90     90     90     95 
```

Initialize Risk Column

```r
# initialize empty risk column
air_quality$risk <- NA

# preview updated dataset
summary(air_quality)
```

```
      day            pm25          season       risk        
 Min.   :  1   Min.   : 2.20   winter:90   Mode:logical  
 1st Qu.: 92   1st Qu.:15.80   spring:90   NA's:365      
 Median :183   Median :27.20   summer:90                 
 Mean   :183   Mean   :26.96   fall  :95                 
 3rd Qu.:274   3rd Qu.:39.10                             
 Max.   :365   Max.   :50.00                             
```

Part 2: Classify Risk with a For Loop

Risk Classification Rules

- **good:** PM2.5 below 12 µg/m³
- **moderate:** PM2.5 between 12 and 35.4 µg/m³
- **unhealthy:** PM2.5 above 35.4 µg/m³

Use a For Loop to Fill Risk Column

```r
# initialize streak trackers
current_streak <- 0
max_unhealthy_streak <- 0

# loop through each day
for (i in 1:nrow(air_quality)) {

  # today's PM2.5 value
  today_pm25 <- air_quality$pm25[i]

  # classify risk category
  if (today_pm25 < 12) {
    air_quality$risk[i] <- "good"

  } else if (today_pm25 <= 35.4) {
    air_quality$risk[i] <- "moderate"

  } else {
    air_quality$risk[i] <- "unhealthy"
  }

  # check unhealthy streak
   if (air_quality$risk[i] == "unhealthy") {

    # continue streak if previous day was unhealthy
    if (i > 1 && air_quality$risk[i - 1] == "unhealthy") {
      current_streak <- current_streak + 1

    } else {
      current_streak <- 1
    }

  } else {
    current_streak <- 0
  }

  # update maximum streak if needed
  if (current_streak > max_unhealthy_streak) {
    max_unhealthy_streak <- current_streak
  }
}
```

```r
# risk category counts
table(air_quality$risk)
```

```

     good  moderate unhealthy 
       70       182       113 
```

```r
# print longest streak
cat("Longest unhealthy streak:", max_unhealthy_streak, "days")
```

```
Longest unhealthy streak: 3 days
```

Part 3: Seasonal Summary with a For Loop

Count Unhealthy Days by Season

```r
# create vector to store unhealthy counts
unhealthy_counts <- c(
  winter = 0,
  spring = 0,
  summer = 0,
  fall = 0
)

# loop through every day
for (i in 1:nrow(air_quality)) {

  # today's season and risk
  today_season <- air_quality$season[i]
  today_risk <- air_quality$risk[i]

  # if unhealthy, add to count
  if (today_risk == "unhealthy") {
    unhealthy_counts[today_season] <- unhealthy_counts[today_season] + 1
  }
}

# display counts
unhealthy_counts
```

```
winter spring summer   fall
    30     23     26     34
```

Convert Seasonal Summary to Data Frame

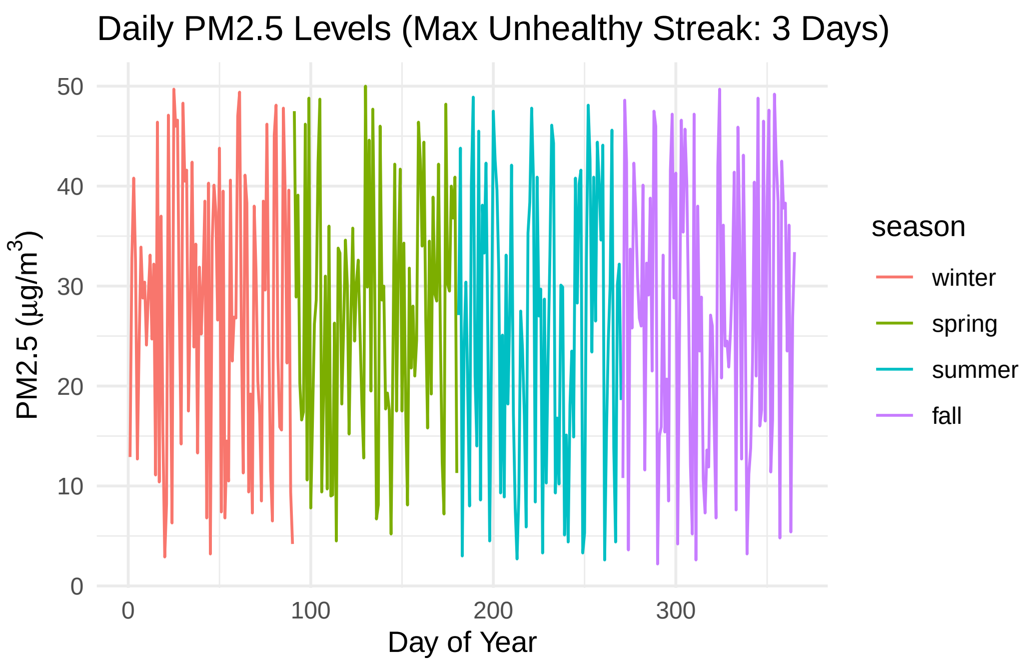
```
# convert counts to data frame
unhealthy_summary <- data.frame(
  season = names(unhealthy_counts),
  unhealthy_days = as.numeric(unhealthy_counts)
)

# display summary table
unhealthy_summary
```

```
  season unhealthy_days
1 winter             30
2 spring             23
3 summer             26
4   fall             34
```

Part 4: Visualize and Report

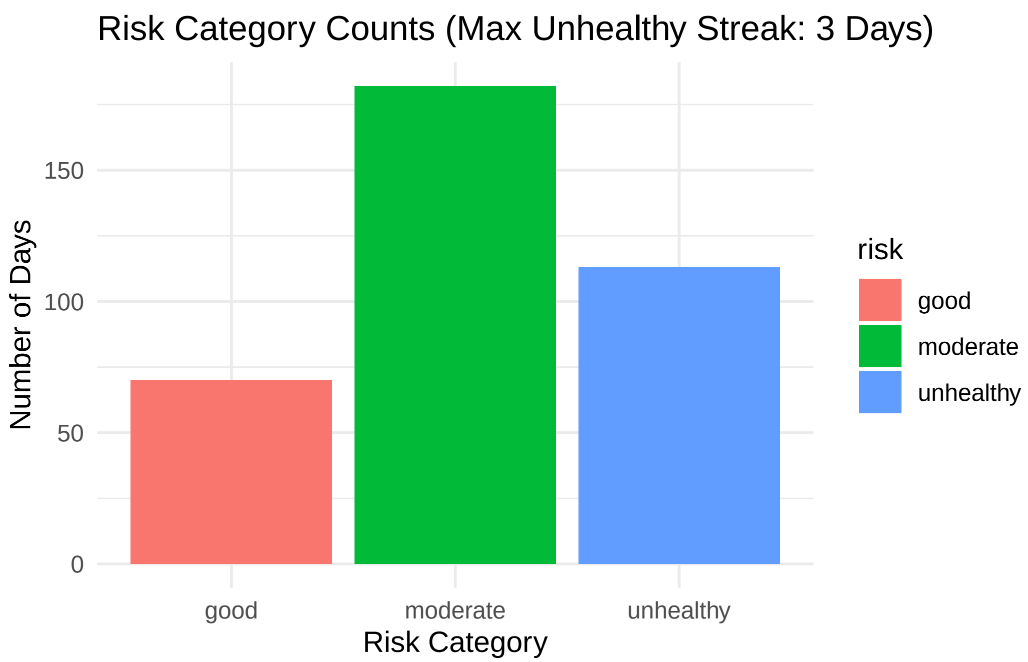
Plot 1: PM2.5 Across the Year

```
# create title using sprintf
plot_title <- sprintf(
  "Daily PM2.5 Levels (Max Unhealthy Streak: %d Days)",
  max_unhealthy_streak
)

# create time series plot
ggplot(air_quality, aes(x = day, y = pm25, color = season)) +
  geom_line() +
  labs(
    title = plot_title,
    x = "Day of Year",
    y = expression("PM2.5 (" * mu * "g/m"^3 * ")")
  ) +
  theme_minimal()
```



Plot 2: Count of Risk Categories

```
# create title using sprintf
bar_title <- sprintf(
  "Risk Category Counts (Max Unhealthy Streak: %d Days)",
  max_unhealthy_streak
)

# create bar chart
ggplot(air_quality, aes(x = risk, fill = risk)) +
  geom_bar() +
  labs(
    title = bar_title,
    x = "Risk Category",
    y = "Number of Days"
  ) +
  theme_minimal()
```



Management Summary

This simulated dataset represents one year of PM2.5 air quality readings at an urban monitoring station. Most days fell within the moderate air quality category, while a substantial number of days were classified as unhealthy. The analysis also identified multiple periods of consecutive unhealthy air quality conditions, with the longest streak lasting 3 consecutive days. Seasonal patterns showed that some seasons experienced more unhealthy days than others, which could reflect seasonal wildfire activity, increased transportation emissions, or weather conditions that trap pollutants near the surface. From a public health and management perspective, these findings suggest that continued air quality monitoring is important for identifying periods of elevated risk. Agencies may benefit from implementing public alert systems, especially during seasons with higher pollution levels, and considering policies aimed at reducing emissions from transportation and industrial sources. Continued monitoring and mitigation efforts could help reduce health risks for sensitive populations such as children, older adults, and individuals with respiratory illnesses.